

# airbnbprediction

April 18, 2025

## TASK 1: Data Exploration and Preprocessing

```
[1]: import pandas as pd

# Load dataset
df = pd.read_csv("Airbnb_data.csv")

# Display basic info
print(" Dataset Info:")
print(df.info())

# First 5 rows
print("\n First few rows:")
print(df.head())

# Missing values
print("\n Missing values:")
print(df.isnull().sum())

# Summary stats
print("\n Summary statistics:")
print(df.describe(include='all'))
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 74111 entries, 0 to 74110
Data columns (total 29 columns):
#   Column                Non-Null Count  Dtype
---  -
0   id                    74111 non-null  int64
1   log_price             74111 non-null  float64
2   property_type         74111 non-null  object
3   room_type             74111 non-null  object
4   amenities             74111 non-null  object
5   accommodates          74111 non-null  int64
6   bathrooms             73911 non-null  float64
7   bed_type              74111 non-null  object
8   cancellation_policy   74111 non-null  object
```

|    |                        |       |          |         |
|----|------------------------|-------|----------|---------|
| 9  | cleaning_fee           | 74111 | non-null | bool    |
| 10 | city                   | 74111 | non-null | object  |
| 11 | description            | 74111 | non-null | object  |
| 12 | first_review           | 58247 | non-null | object  |
| 13 | host_has_profile_pic   | 73923 | non-null | object  |
| 14 | host_identity_verified | 73923 | non-null | object  |
| 15 | host_response_rate     | 55812 | non-null | object  |
| 16 | host_since             | 73923 | non-null | object  |
| 17 | instant_bookable       | 74111 | non-null | object  |
| 18 | last_review            | 58284 | non-null | object  |
| 19 | latitude               | 74111 | non-null | float64 |
| 20 | longitude              | 74111 | non-null | float64 |
| 21 | name                   | 74111 | non-null | object  |
| 22 | neighbourhood          | 67239 | non-null | object  |
| 23 | number_of_reviews      | 74111 | non-null | int64   |
| 24 | review_scores_rating   | 57389 | non-null | float64 |
| 25 | thumbnail_url          | 65895 | non-null | object  |
| 26 | zipcode                | 73143 | non-null | object  |
| 27 | bedrooms               | 74020 | non-null | float64 |
| 28 | beds                   | 73980 | non-null | float64 |

dtypes: bool(1), float64(7), int64(3), object(18)

memory usage: 15.9+ MB

None

First few rows:

|   | id       | log_price | property_type | room_type       | \ |
|---|----------|-----------|---------------|-----------------|---|
| 0 | 6901257  | 5.010635  | Apartment     | Entire home/apt |   |
| 1 | 6304928  | 5.129899  | Apartment     | Entire home/apt |   |
| 2 | 7919400  | 4.976734  | Apartment     | Entire home/apt |   |
| 3 | 13418779 | 6.620073  | House         | Entire home/apt |   |
| 4 | 3808709  | 4.744932  | Apartment     | Entire home/apt |   |

|   | amenities                                         | accommodates | bathrooms | \ |
|---|---------------------------------------------------|--------------|-----------|---|
| 0 | {"Wireless Internet","Air conditioning",Kitche... | 3            | 1.0       |   |
| 1 | {"Wireless Internet","Air conditioning",Kitche... | 7            | 1.0       |   |
| 2 | {TV,"Cable TV","Wireless Internet","Air condit... | 5            | 1.0       |   |
| 3 | {TV,"Cable TV",Internet,"Wireless Internet",Ki... | 4            | 1.0       |   |
| 4 | {TV,Internet,"Wireless Internet","Air conditio... | 2            | 1.0       |   |

|   | bed_type | cancellation_policy | cleaning_fee | ... | latitude  | longitude   | \ |
|---|----------|---------------------|--------------|-----|-----------|-------------|---|
| 0 | Real Bed | strict              | True         | ... | 40.696524 | -73.991617  |   |
| 1 | Real Bed | strict              | True         | ... | 40.766115 | -73.989040  |   |
| 2 | Real Bed | moderate            | True         | ... | 40.808110 | -73.943756  |   |
| 3 | Real Bed | flexible            | True         | ... | 37.772004 | -122.431619 |   |
| 4 | Real Bed | moderate            | True         | ... | 38.925627 | -77.034596  |   |

|   | name                           | neighbourhood    | \ |
|---|--------------------------------|------------------|---|
| 0 | Beautiful brownstone 1-bedroom | Brooklyn Heights |   |

|   |                                          |                  |
|---|------------------------------------------|------------------|
| 1 | Superb 3BR Apt Located Near Times Square | Hell's Kitchen   |
| 2 | The Garden Oasis                         | Harlem           |
| 3 | Beautiful Flat in the Heart of SF!       | Lower Haight     |
| 4 | Great studio in midtown DC               | Columbia Heights |

|   | number_of_reviews | review_scores_rating \ |
|---|-------------------|------------------------|
| 0 | 2                 | 100.0                  |
| 1 | 6                 | 93.0                   |
| 2 | 10                | 92.0                   |
| 3 | 0                 | NaN                    |
| 4 | 4                 | 40.0                   |

|   | thumbnail_url                                                                                                     | zipcode | bedrooms | beds |
|---|-------------------------------------------------------------------------------------------------------------------|---------|----------|------|
| 0 | <a href="https://a0.muscache.com/im/pictures/6d7cbbf7-c...">https://a0.muscache.com/im/pictures/6d7cbbf7-c...</a> | 11201   | 1.0      | 1.0  |
| 1 | <a href="https://a0.muscache.com/im/pictures/348a55fe-4...">https://a0.muscache.com/im/pictures/348a55fe-4...</a> | 10019   | 3.0      | 3.0  |
| 2 | <a href="https://a0.muscache.com/im/pictures/6fae5362-9...">https://a0.muscache.com/im/pictures/6fae5362-9...</a> | 10027   | 1.0      | 3.0  |
| 3 | <a href="https://a0.muscache.com/im/pictures/72208dad-9...">https://a0.muscache.com/im/pictures/72208dad-9...</a> | 94117   | 2.0      | 2.0  |
| 4 | NaN                                                                                                               | 20009   | 0.0      | 1.0  |

[5 rows x 29 columns]

Missing values:

|                        |       |
|------------------------|-------|
| id                     | 0     |
| log_price              | 0     |
| property_type          | 0     |
| room_type              | 0     |
| amenities              | 0     |
| accommodates           | 0     |
| bathrooms              | 200   |
| bed_type               | 0     |
| cancellation_policy    | 0     |
| cleaning_fee           | 0     |
| city                   | 0     |
| description            | 0     |
| first_review           | 15864 |
| host_has_profile_pic   | 188   |
| host_identity_verified | 188   |
| host_response_rate     | 18299 |
| host_since             | 188   |
| instant_bookable       | 0     |
| last_review            | 15827 |
| latitude               | 0     |
| longitude              | 0     |
| name                   | 0     |
| neighbourhood          | 6872  |
| number_of_reviews      | 0     |
| review_scores_rating   | 16722 |
| thumbnail_url          | 8216  |

```

zipcode          968
bedrooms         91
beds            131
dtype: int64

```

Summary statistics:

|        | id           | log_price    | property_type | room_type       | amenities | \ |
|--------|--------------|--------------|---------------|-----------------|-----------|---|
| count  | 7.411100e+04 | 74111.000000 | 74111         | 74111           | 74111     |   |
| unique | NaN          | NaN          | 35            | 3               | 67122     |   |
| top    | NaN          | NaN          | Apartment     | Entire home/apt | {}        |   |
| freq   | NaN          | NaN          | 49003         | 41310           | 586       |   |
| mean   | 1.126662e+07 | 4.782069     | NaN           | NaN             | NaN       |   |
| std    | 6.081735e+06 | 0.717394     | NaN           | NaN             | NaN       |   |
| min    | 3.440000e+02 | 0.000000     | NaN           | NaN             | NaN       |   |
| 25%    | 6.261964e+06 | 4.317488     | NaN           | NaN             | NaN       |   |
| 50%    | 1.225415e+07 | 4.709530     | NaN           | NaN             | NaN       |   |
| 75%    | 1.640226e+07 | 5.220356     | NaN           | NaN             | NaN       |   |
| max    | 2.123090e+07 | 7.600402     | NaN           | NaN             | NaN       |   |

|        | accommodates | bathrooms    | bed_type | cancellation_policy | cleaning_fee | \ |
|--------|--------------|--------------|----------|---------------------|--------------|---|
| count  | 74111.000000 | 73911.000000 | 74111    | 74111               | 74111        |   |
| unique | NaN          | NaN          | 5        | 5                   | 2            |   |
| top    | NaN          | NaN          | Real Bed | strict              | True         |   |
| freq   | NaN          | NaN          | 72028    | 32374               | 54403        |   |
| mean   | 3.155146     | 1.235263     | NaN      | NaN                 | NaN          |   |
| std    | 2.153589     | 0.582044     | NaN      | NaN                 | NaN          |   |
| min    | 1.000000     | 0.000000     | NaN      | NaN                 | NaN          |   |
| 25%    | 2.000000     | 1.000000     | NaN      | NaN                 | NaN          |   |
| 50%    | 2.000000     | 1.000000     | NaN      | NaN                 | NaN          |   |
| 75%    | 4.000000     | 1.000000     | NaN      | NaN                 | NaN          |   |
| max    | 16.000000    | 8.000000     | NaN      | NaN                 | NaN          |   |

|        | ... | latitude     | longitude    | name   | neighbourhood | \ |
|--------|-----|--------------|--------------|--------|---------------|---|
| count  | ... | 74111.000000 | 74111.000000 | 74111  | 67239         |   |
| unique | ... | NaN          | NaN          | 73331  | 619           |   |
| top    | ... | NaN          | NaN          | #NAME? | Williamsburg  |   |
| freq   | ... | NaN          | NaN          | 10     | 2862          |   |
| mean   | ... | 38.445958    | -92.397525   | NaN    | NaN           |   |
| std    | ... | 3.080167     | 21.705322    | NaN    | NaN           |   |
| min    | ... | 33.338905    | -122.511500  | NaN    | NaN           |   |
| 25%    | ... | 34.127908    | -118.342374  | NaN    | NaN           |   |
| 50%    | ... | 40.662138    | -76.996965   | NaN    | NaN           |   |
| 75%    | ... | 40.746096    | -73.954660   | NaN    | NaN           |   |
| max    | ... | 42.390437    | -70.985047   | NaN    | NaN           |   |

|        | number_of_reviews | review_scores_rating | \ |
|--------|-------------------|----------------------|---|
| count  | 74111.000000      | 57389.000000         |   |
| unique | NaN               | NaN                  |   |

|      |            |            |
|------|------------|------------|
| top  | NaN        | NaN        |
| freq | NaN        | NaN        |
| mean | 20.900568  | 94.067365  |
| std  | 37.828641  | 7.836556   |
| min  | 0.000000   | 20.000000  |
| 25%  | 1.000000   | 92.000000  |
| 50%  | 6.000000   | 96.000000  |
| 75%  | 23.000000  | 100.000000 |
| max  | 605.000000 | 100.000000 |

|        |                                                   |         |   |
|--------|---------------------------------------------------|---------|---|
|        | thumbnail_url                                     | zipcode | \ |
| count  | 65895                                             | 73143   |   |
| unique | 65883                                             | 668     |   |
| top    | https://a0.muscache.com/im/pictures/70087089/b... | 11211   |   |
| freq   | 3                                                 | 1618    |   |
| mean   | NaN                                               | NaN     |   |
| std    | NaN                                               | NaN     |   |
| min    | NaN                                               | NaN     |   |
| 25%    | NaN                                               | NaN     |   |
| 50%    | NaN                                               | NaN     |   |
| 75%    | NaN                                               | NaN     |   |
| max    | NaN                                               | NaN     |   |

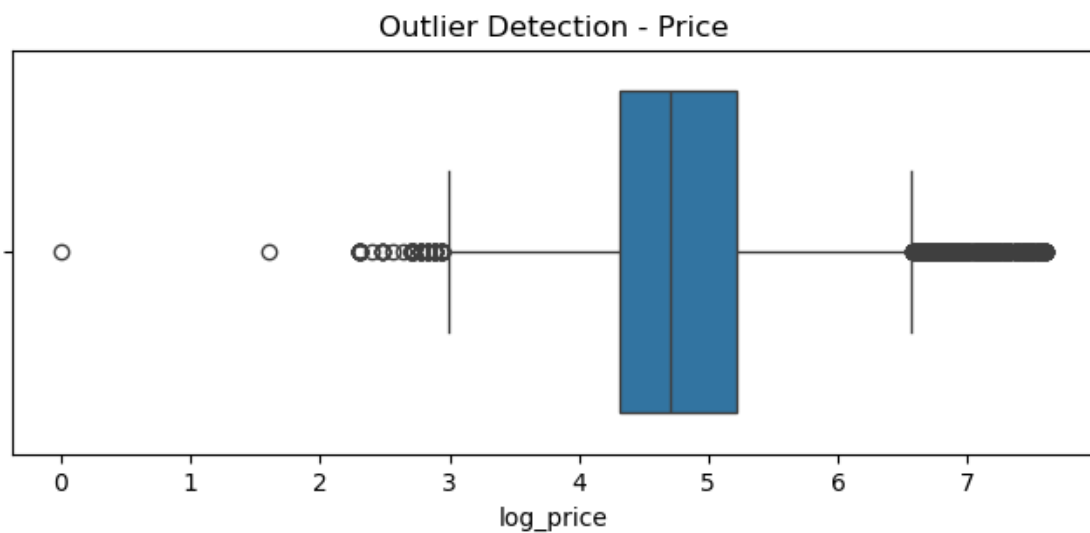
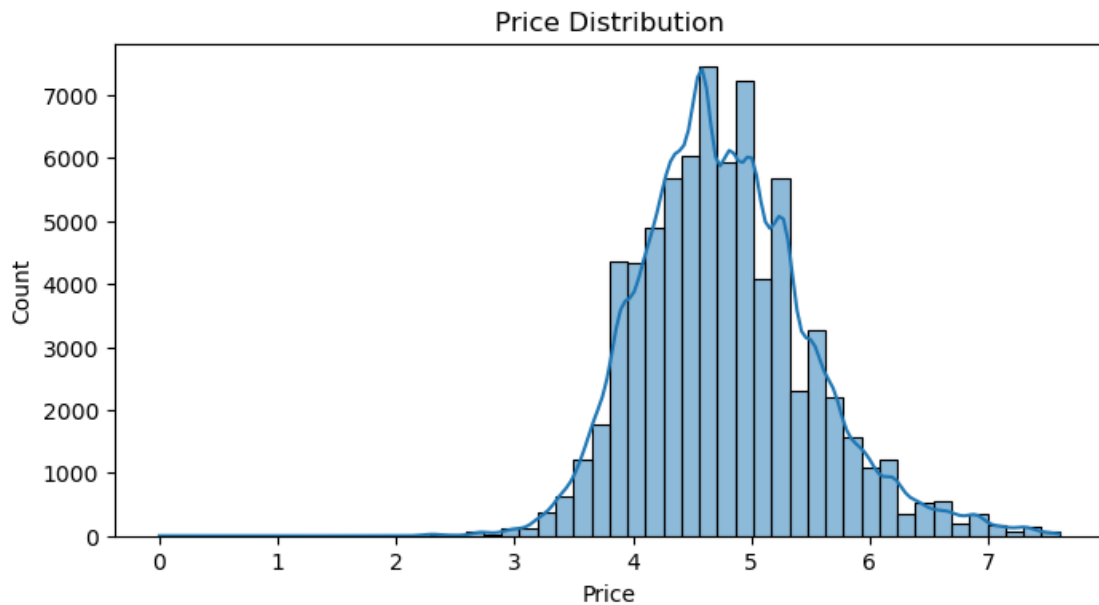
|        |              |              |
|--------|--------------|--------------|
|        | bedrooms     | beds         |
| count  | 74020.000000 | 73980.000000 |
| unique | NaN          | NaN          |
| top    | NaN          | NaN          |
| freq   | NaN          | NaN          |
| mean   | 1.265793     | 1.710868     |
| std    | 0.852143     | 1.254142     |
| min    | 0.000000     | 0.000000     |
| 25%    | 1.000000     | 1.000000     |
| 50%    | 1.000000     | 1.000000     |
| 75%    | 1.000000     | 2.000000     |
| max    | 10.000000    | 18.000000    |

[11 rows x 29 columns]

```
[5]: import seaborn as sns
import matplotlib.pyplot as plt

# Price distribution
plt.figure(figsize=(8, 4))
sns.histplot(df['log_price'], bins=50, kde=True)
plt.title("Price Distribution")
plt.xlabel("Price")
plt.show()
```

```
# Boxplot to spot outliers
plt.figure(figsize=(8, 3))
sns.boxplot(x=df['log_price'])
plt.title("Outlier Detection - Price")
plt.show()
```



```
[7]: # Drop rows with missing price or key fields
df_clean = df.dropna(subset=['log_price', 'room_type', 'property_type'])

# Remove extreme outliers
df_clean = df_clean[df_clean['log_price'] < df_clean['log_price'].quantile(0.
↪99)]

# Fill missing numeric columns with median
numeric_cols = df_clean.select_dtypes(include=['float64', 'int64']).columns
df_clean[numeric_cols] = df_clean[numeric_cols].fillna(df_clean[numeric_cols].
↪median())
```

## TASK 2: Model Development

```
[9]: from sklearn.preprocessing import LabelEncoder

# Make a copy of the cleaned dataframe
df_label = df_clean.copy()

# Identify categorical columns
cat_cols = df_label.select_dtypes(include=['object']).columns.tolist()

# Excluded non-feature columns
cat_cols = [col for col in cat_cols if col not in ['log_price', 'id', 'name',
↪'description']] # adjust as needed

# Apply label encoding to each categorical column
le = LabelEncoder()
for col in cat_cols:
    df_label[col] = df_label[col].astype(str) # ensure all entries are strings
    df_label[col] = le.fit_transform(df_label[col])

# Drop raw text columns that can't be converted to numbers directly
text_columns = ['name', 'description', 'neighborhood',
↪'amenities', 'thumbnail_url']
for col in text_columns:
    if col in df_label.columns:
        df_label.drop(col, axis=1, inplace=True)
```

```
[11]: from sklearn.model_selection import train_test_split

# Define features and target
X = df_label.drop('log_price', axis=1)
y = df_label['log_price']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
↪random_state=42)
```

```
print(f" Feature matrix shape: {X.shape}")
feature_names = X_train.columns.tolist()
```

Feature matrix shape: (73369, 24)

```
[14]: from sklearn.linear_model import LinearRegression
      from sklearn.ensemble import RandomForestRegressor
      from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

      # Initialize models
      lr_model = LinearRegression()
      rf_model = RandomForestRegressor(n_estimators=100, random_state=42)

      # Train models
      lr_model.fit(X_train, y_train)
      rf_model.fit(X_train, y_train)
```

[14]: RandomForestRegressor(random\_state=42)

```
[15]: # Predictions
      lr_preds = lr_model.predict(X_test)
      rf_preds = rf_model.predict(X_test)
```

### TASK 3: Model Evaluation

```
[16]: import numpy as np
      def evaluate_model(name, y_true, y_pred):
          print(f"\n {name} Evaluation")
          print(f"MAE (Mean Absolute Error): {mean_absolute_error(y_true, y_pred):.2f}")
          print(f"MSE (Mean Squared Error): {mean_squared_error(y_true, y_pred):.2f}")
          print(f"RMSE (Root MSE): {np.sqrt(mean_squared_error(y_true, y_pred)):.2f}")
          print(f"R2 Score: {r2_score(y_true, y_pred):.4f}")

      # Evaluate both models
      evaluate_model("Linear Regression", y_test, lr_preds)
      evaluate_model("Random Forest", y_test, rf_preds)
```

Linear Regression Evaluation  
MAE (Mean Absolute Error): 0.35  
MSE (Mean Squared Error): 0.21  
RMSE (Root MSE): 0.46  
R<sup>2</sup> Score: 0.5423



Random Forest Evaluation  
MAE (Mean Absolute Error): 0.28  
MSE (Mean Squared Error): 0.14  
RMSE (Root MSE): 0.38  
R<sup>2</sup> Score: 0.6984

```
[18]: import matplotlib.pyplot as plt
import seaborn as sns

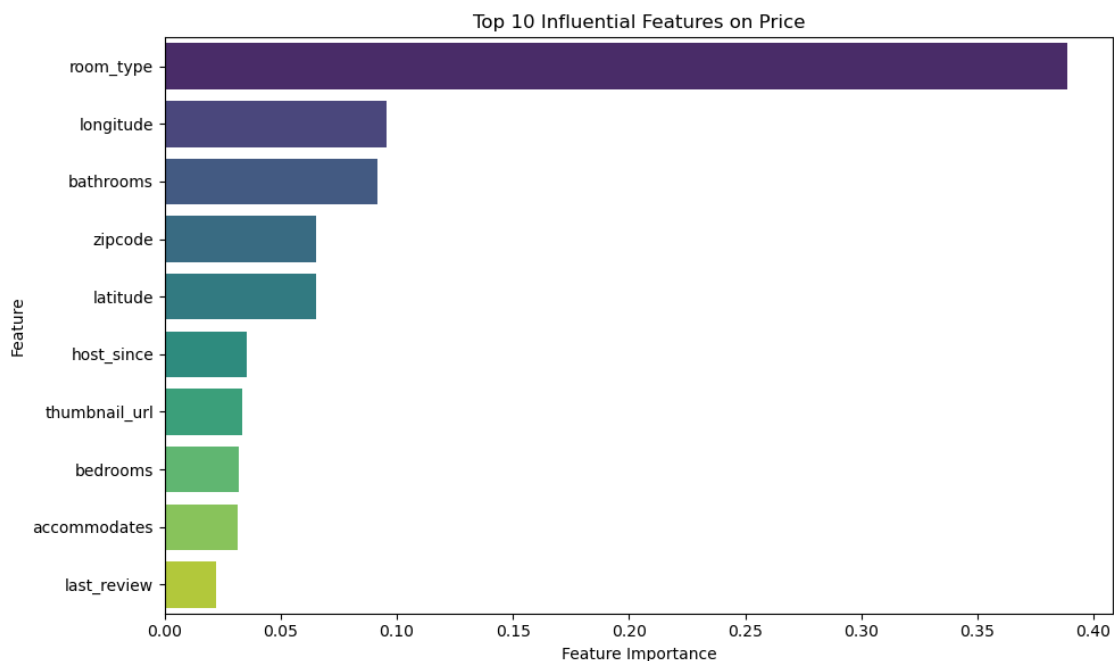
feat_importances = pd.Series(rf_model.feature_importances_, index=X.columns).
    ↪sort_values(ascending=False)[:10]

plt.figure(figsize=(10, 6))
sns.barplot(x=feat_importances, y=feat_importances.index, palette='viridis')
plt.title("Top 10 Influential Features on Price")
plt.xlabel("Feature Importance")
plt.ylabel("Feature")
plt.tight_layout()
plt.show()
```

C:\Users\Vedant Jadhav\AppData\Local\Temp\ipykernel\_26596\2586554865.py:7:  
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x=feat_importances, y=feat_importances.index, palette='viridis')
```



VIDEO LINK: <https://drive.google.com/file/d/1qaBleDCP1iMjMNbUPTfwII0f6kw66g1/view?usp=drivesdk>